Input Form (IF)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: *MMLU* | ***Context:*** *North India Competitive Exam Aspirants (Hindi-Medium)*

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MMLU is entirely English-language, text-only, with a standardised English preamble for prompts [Q44] and English-only normalised input format for UnifiedQA [Q91]. Dataset metadata confirms 'languages: [en]' and monolingual content; no Devanagari script or Hindi vocabulary appears in any of the ~200 sampled items. The deployment requires Hindi-dominant interaction in Devanagari script with at most ~10% English code-mixing per elicitation Q4. The MCQ structural format (4 options A/B/C/D) is a partial structural match to UPSC/SSC exam formats, but this is overridden by total language and script mismatch. IndicEval [WEB-16] empirically demonstrates a large Hindi-vs-English performance gap (LLaMA 39.53% on UPSC Hindi Zero-Shot vs. ~84.88% Gemini on UPSC English CoT), confirming that English-medium MMLU scores cannot predict Hindi-medium performance.

ID	Evidence
Q44	“We feed GPT-3 prompts like that shown in Figure 1a. We begin each prompt with "The following are multiple choice questions (with answers) about [subject]." For zero-shot evaluation, we append the question to the prompt. For few-shot evaluation, we add up to 5 demonstration examples with answers to the prompt before appending the question. All prompts end with "Answer: ". The model then produces probabilities for the tokens "A," "B," "C," and "D," and we treat the highest probability option as the prediction. For consistent evaluation, we create a dev set with 5 fixed few-shot examples for each subject.” (p.6)
Q91	“UnifiedQA's input format is of the form QUESTION1 \n (A) CHOICE1 (B) CHOICE2 (C) CHOICE3 (D) CHOICE4</s> where questions and choices are normalized and made lowercase.” (p.12)
Q4	“However, on every one of the 57 tasks, the best models still need substantial improvements before they can reach expert-level accuracy.” (p.1)
WEB-16	IndicEval documents marked accuracy drops in Hindi vs. English (LLaMA 39.53% UPSC Hindi Zero-Shot) (https://arxiv.org/abs/2602.16467)

Strengths

- Four-option MCQ structure (A/B/C/D) [Q44, Q83] is a structural match to UPSC Prelims, SSC CGL, and banking exam formats [DATASET strength 1].
- Text-only modality is appropriate for the deployment's text-based MCQ context (per elicitation deployment description, mobile/enterprise application is text-driven).

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Q44	“We feed GPT-3 prompts like that shown in Figure 1a. We begin each prompt with "The following are multiple choice questions (with answers) about [subject]." For zero-shot evaluation, we append the question to the prompt. For few-shot evaluation, we add up to 5 demonstration examples with answers to the prompt before appending the question. All prompts end with "Answer: ". The model then produces probabilities for the tokens "A," "B," "C," and "D," and we treat the highest probability option as the prediction. For consistent evaluation, we create a dev set with 5 fixed few-shot examples for each subject.” (p.6)
Q83	“Models are fine-tuned to predict one of four classes using the UnifiedQA MCQ questions and using our dev+val set. We test on our multitask test set.” (p.12)

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Severe mismatch — MMLU input distribution is 100% English text [Q44, Q91]; deployment input distribution is Hindi (Devanagari) with up to ~10% English code-mixing.

IF-2: Regional infrastructure does support Devanagari rendering (Gboard, SwiftKey Hindi available) but the benchmark provides no Devanagari content to evaluate; Hindi tokenization quality is a documented live deployment risk [WEB-1].

IF-3: Form-relevant differences: MMLU uses no Devanagari script, no Hindi morphology, no code-mixed Hinglish; mathematics items use US customary units (miles, feet) rather than SI units used in Indian exams [DATASET-D25, D28].

IF-4: External validity is severely harmed: MMLU accuracy on English inputs provides no evidence about model performance on Hindi-medium inputs; IndicEval [WEB-16] documents catastrophic multilingual degradation.

ID	Evidence
Q44	"We feed GPT-3 prompts like that shown in Figure 1a. We begin each prompt with "The following are multiple choice questions (with answers) about [subject]." For zero-shot evaluation, we append the question to the prompt. For few-shot evaluation, we add up to 5 demonstration examples with answers to the prompt before appending the question. All prompts end with "Answer: ". The model then produces probabilities for the tokens "A," "B," "C," and "D," and we treat the highest probability option as the prediction. For consistent evaluation, we create a dev set with 5 fixed few-shot examples for each subject." (p.6)
Q91	"UnifiedQA's input format is of the form QUESTION1 \n (A) CHOICE1 (B) CHOICE2 (C) CHOICE3 (D) CHOICE4</s> where questions and choices are normalized and made lowercase." (p.12)
WEB-16	IndicEval documents marked accuracy drops in Hindi vs. English (LLaMA 39.53% UPSC Hindi Zero-Shot) (https://arxiv.org/abs/2602.16467)
WEB-1	Devanagari tokenization quality is a live deployment risk; even models trained on 20B Hindi tokens hallucinate or mix in English (https://arxiv.org/abs/2410.14815)
Q59	"While text is capable of conveying an enormous number of concepts about the world, many important concepts are conveyed mainly through other modalities, such as images, audio, and physical interaction (Bisk et al., 2020). Existing large-scale NLP models, such as GPT-3, do not incorporate multimodal information, so we design our benchmark to capture a diverse array of tasks in a text-only format." (p.7)
D25	If a freight train travels at a speed of 20 miles per hour for 6 hours, how far will it travel? ... 120 miles — Uses miles (US customary unit) — Indian exams use km; creates construct-irrelevant variance
D28	A total of x feet of fencing is to form three sides of a level rectangular yard. What is the maximum possible area of the yard, in terms of x? — Uses feet (US customary unit) — minor but consistent US measurement convention

Information Gaps

- Quality of community Hindi MMLU translations (OpenAI MMMLU, IndicMMLU-Pro) is documented in web sources but not directly assessed here.

Requires Expert Verification

- Hindi linguistic review of any translated MMLU variant for register appropriateness (Manak Hindi vs. literal translation) for exam-prep context.

Remediation

Gap 1: 100% English input with no Devanagari script; deployment requires Hindi-dominant input.

Recommendation: Adopt IndicEval [WEB-16] for direct UPSC-Hindi performance measurement and IndicMMLU-Pro [WEB-18] for broader Hindi-translation assessment. Treat any English-MMLU score as upper-bound only; expect substantial Hindi-medium degradation.

ID	Evidence
WEB-16	IndicEval documents marked accuracy drops in Hindi vs. English (LLaMA 39.53% UPSC Hindi Zero-Shot) (https://arxiv.org/abs/2602.16467)
WEB-18	https://arxiv.org/abs/2501.15747